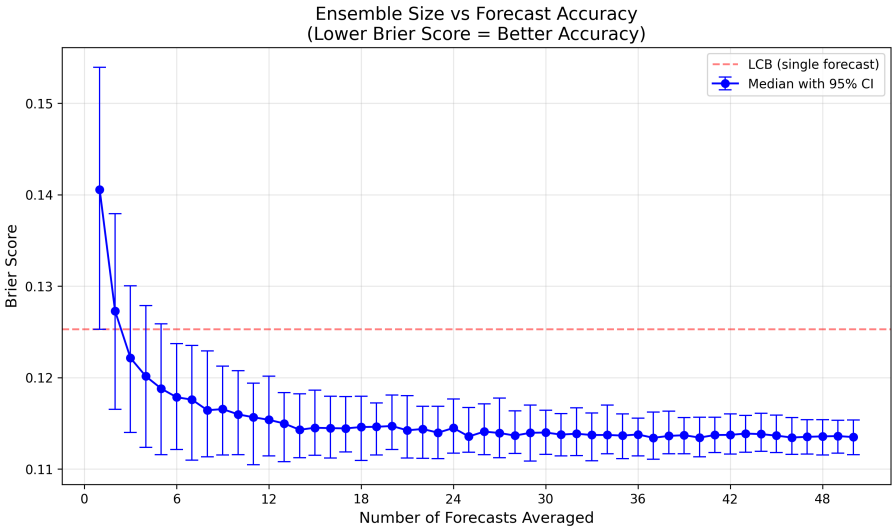

Figure 3: The Brier score induced at various ensemble sizes. Point estimates and 95% confidence intervals are generated via bootstrap resampling from a set of 50 forecasts per question. The dashed line indicates the lower confidence bound for a single forecast.

Synthesis Method	**Brier score**
Simple Mean	0.1140
Trimmed Mean	0.1142
Median	0.1138
Single Forecast	0.1182

Table 9: Evaluating different approaches to ensembling multiple forecasts on FB-7-21. The mean, trimmed mean and median aggregation over 10 forecasts all perform substantially better than a single randomly chosen forecast.

Figure 3 also suggests diminishing returns beyond a handful of independent forecasts; we adopt 10 forecasts per question as a conservative standard in this work. This result also illustrates that sampling only a single forecast for each question yields highly variable performance—even when averaged over the 1610 questions in this evaluation set—due to the inherent randomness in LLM outputs. This variability is itself an undesirable property for a forecaster, and suggests that some form of ensembling is critical for internal validity. This kind of ensembling has thus far not been common in the LLM forecasting literature[16], and suggests that the common practice of quantifying uncertainty *only over forecasting questions themselves*—rather than also considering uncertainty over LLM outputs—dramatically understates the true variability of LLM forecasting performance.

In Table 9, we further demonstrate that minor variations of this approach, like taking the median of 10 forecasts, or taking the trimmed mean (by discarding the smallest and largest of the 10 forecasts), perform comparably to simple averaging. We include a reference baseline which samples a single individual forecast for each question uniformly at random.

16. One exception is Schoenegger et al. (2024), though this work is motivated by the hypothesis that ensembling *different* LLM-based forecasters will improve performance.

The results above suggest that averaging multiple independent forecasts is critical for forecasting performance and cannot be meaningfully improved by minor variations thereof.

7.2 Supervisor Agents Resolve Disagreements

We now turn to more sophisticated ensembling methods, which utilize an LLM-based supervisor to synthesize multiple independent forecasts. This supervisor can be viewed as a kind of generative verifier or LLM-as-a-judge (Li et al., 2024; Zhang et al., 2024) which aggregates the 10 forecasts generated by each agent into a final probability.

We consider three different supervisor agents, as well as the same no aggregation baseline described in Section 7.1. The first, which we call the "best of k" supervisor, is provided with 10 forecasts and asked to reason over them to output the best of these forecasts. Importantly, the supervisor is explicitly constrained to output exactly one of the provided forecasts. The second, which we call the "non-agentic supervisor," uses the method proposed in Halawi et al. (2024) to ensemble multiple forecasts into a final forecast. Unlike the best of k method, this method is not constrained to output one of the input forecasts. Instead, the model simply uses these as inputs to reason toward a stronger forecast. Finally, we propose the "agentic supervisor," which proceeds in three steps. First, the supervisor model identifies disagreements among the 10 input forecasts. Second, the supervisor generates and executes a series of search queries intended to resolve these disagreements (e.g., to resolve disagreement over the accurate base rate for an event). Finally, the supervisor generates an updated forecast along with its confidence that the forecast was updated in the correct direction ("high," "medium" or "low"). High confidence updates are used in place of the simple mean, while medium or low confidence updates are discarded.

Synthesis Method	Brier score	Top@3	Worst@3	Outperform (%)
Agentic Supervisor	**0.1125**	**33.9%**	**4.6%**	**3.8%**
Non-agentic Supervisor	0.1168	7.8%	6.2%	0.6%
Best of k (LLM)	0.1191	29.7%	7.2%	0.0%
None	0.1199			

Table 10: Comparison of different synthesis strategies on FB-7-21. **Top@3**: fraction of aggregated forecasts which are at least as accurate as the three best individual forecasts; **Worst@3**: fraction of aggregated forecasts which are no more accurate than the three worst individual forecasts; **Outperform (%)**: percentage of aggregate forecasts beating all individual forecasts. Agentic supervision yields the best overall performance.

As shown in Table 10, the agentic supervisor achieves the best performance among the synthesis methods. The best of k approach, which selects a single forecast from the ensemble to serve as the final prediction, performs poorly. It never selects a forecast that is better than any of the candidate forecasts by design, and it happens to pick among the worst of the candidate forecasts with a striking 7.2% frequency. The non-agentic supervisor, in which a language model performs the aggregation, shows modest improvement over

the best of k method. However, this method produces quality forecasts (top@3 of 7.8%) relatively rarely. In contrast, the agentic supervisor substantially outperforms both alternatives. It actively reconciles inconsistencies between individual predictions and performs additional news queries to refine forecast rationales. This synthesis process enables it to achieve the lowest Brier score of 0.1125, highest top@3 accuracy, highest outperformance rate, and lowest worst@3 rate, demonstrating the value of additional search and agentic reasoning when aggregating forecasts.

7.3 Statistical Corrections and Where They Help

Statistical corrections are commonly applied to forecasts to correct for distortions in the forecasted probabilities (Niculescu-Mizil and Caruana, 2005; Baron et al., 2014). In the binary case, Niculescu-Mizil and Caruana note that forecasts often result in probabilities that are clustered away from zero or one, which we observe in our data (Figure 4).

We consider statistical correction functions $f_\theta(p) : [0, 1] \to [0, 1]$ (Neyman and Roughgarden, 2022), which may either have learnable parameters (Niculescu-Mizil and Caruana, 2005) or are deterministic functions (Satopää et al., 2014).

Correction Method	Fixed Parameter	ID Training	OOD Training
Platt Scaling	**0.1076**	**0.1071**	**0.1104**
Log Odds Extremization	0.1085	×	×
Isotonic Regression	×	0.1097	0.1134
OLS	×	0.1119	0.1125
None	0.1140	×	×

Table 11: Evaluating different statistical correction approaches on FB-7-21. The fixed parameter setting sets the scaling parameter to $\sqrt{3}$, as proposed by Neyman and Roughgarden (2022). In distribution (ID) training reports leave-one-out performance with learned scaling parameters. Out of distribution (OOD) training fits the scaling parameters on the benchmark released by Halawi et al. (2024). Platt scaling yields the best performance across these methods.

Table 11 shows that the most beneficial statistical correction methods are Platt scaling (Niculescu-Mizil and Caruana, 2005) and extremization (Neyman and Roughgarden, 2022). Log odds extremization aggregates a set of k forecasts by taking a scaled mean of the log odds of each forecast. The scaling parameter can be learned from data or set to a fixed value a priori. Log odds extremization follows the following formula (Baron et al., 2014):

$$\log \frac{\hat{p}}{1 - \hat{p}} = \frac{d}{n} \sum_{i=1}^{n} \log \frac{p_i}{1 - p_i}, \tag{2}$$

where the aggregate estimate $\hat{p}$ can then be recovered using the sigmoid function. We demonstrate that Equation 2 is mathematically identical to applying Platt scaling to the geometric mean of forecasts with a Platt coefficient of d in Appendix G.